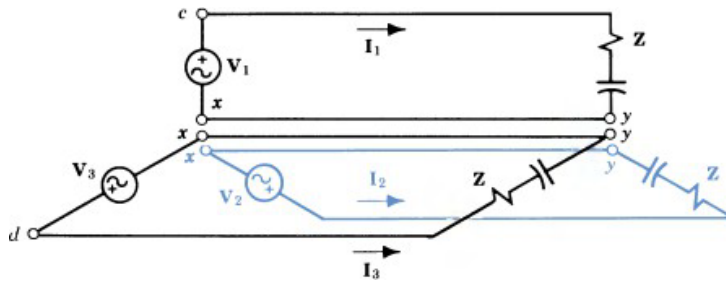


Tutorial – 7

ELL 101: Introduction to Electrical Engineering

- 1) A load consisting of three identical impedances $Z = 10\angle -45^\circ \Omega$ is connected in Δ to a three-phase, 220 V source. Determine phase and line currents.
- 2) Given the circuit with:

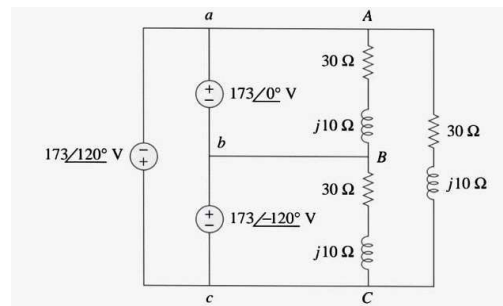


$$V_1 = 240\angle 90^\circ \text{ V} \quad V_3 = 240\angle -150^\circ \text{ V}$$

$$V_2 = 240\angle -30^\circ \text{ V} \quad Z = 40\angle -30^\circ \Omega$$

- (a) Calculate phasors I_1 , I_2 and I_3 .
 - (b) If terminals x are connected together and terminals y are connected together, what will be Current I_{yx} ?
 - (c) Under the condition of part (b), what will be the voltage V_{cd} ?
 - (d) Draw the phasors of V_1 , V_2 , V_3 and I_1 , I_2 , I_3 on a single phasor diagram with all phasors starting from a common point.
- 3) Three equal impedances are connected in Δ across a 240-V, 3- ϕ line. One of the three line- to line voltages is $240\angle 60^\circ \text{ V}$ and one of the phase currents is $6\angle 200^\circ \text{ A}$.
 - (a) Identify the other voltages and phase currents and show all on a phasor diagram (all phasors starting from the same point).
 - (b) Determine the phase impedance in complex form.
 - (c) Determine the three line currents in phasor form
 - 4) Three equal impedances of $20\angle -30^\circ \Omega$ are connected in Δ across a 240-V, 3- ϕ line.
 - (a) Draw a clearly labelled wiring diagram properly oriented.
 - (b) Calculate phase and line currents.
 - (c) Draw a labelled phasor diagram of voltages and currents
 - (d) Calculate the total power.

5) For the following Δ - Δ circuit, calculate the phase and line currents.



6) A Δ -connected source supplies power to a Y-connected load in a three-phase balanced system. Given that the line impedance is $2 + j1 \Omega$ per phase while the load impedance is $6 + j4 \Omega$ per phase, find the magnitude of the line voltage at the load. Assume the source phase voltage $V_{ab} = 208 \angle 0^\circ$ Vrms.

7) Given the following circuit, find the total complex power absorbed by the load.

